

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

24th May 2023

Subject SP6 – Financial Derivatives

Time allowed: 3 Hours 15 Minutes (10.15 – 13.30 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. *The answers are not expected to be any country or jurisdiction specific. However, if Examples/illustrations are required for any answer, the country or jurisdiction from which they are drawn should be mentioned.*
3. *Attempt all questions beginning your answer to each question on a separate sheet.*
4. *Mark allocations are shown in brackets.*
5. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

Q. 1) What is the Longstaff-Schwartz approach to price an American option? Describe the steps involved in pricing an American option using the Longstaff-Schwartz method. **[9]**

Q. 2) Given the market volatility and stronger expected performance on Indian equities, the Chief Information Officer of an active investment management firm is considering changing the asset allocation of INR 100 Cr portfolio as below.

	Current	Future
Indian equity	40%	60%
UK equity	30%	20%
US equity	30%	20%

Currently assets are held in individual equities (instead of overall index exposure) and the fund manager is expected to beat index returns and generate alpha.

- i)** Suggest three approaches which can be used to achieve above portfolio position effective immediately. Compare the advantages, risk, performance free impact, currency impact and other issues if any for these approaches. **(4)**
- ii)** Suggest the approach that firm should adopt with rationale supporting the decision based on above factors. **(2)**
- iii)** The firm has decided to enter into a 3-month future transaction on benchmark index to immediately implement the above portfolio allocation. UK and US index future are linked to their respective currencies. Level of various benchmark index on date of transaction are BSE(India): 60000, S&P (US): 4000, FTSE(UK): 7000, Interest rate INDIA: 8% , UK: 5% , US: 6% annually compounded. Spot rate: USD/INR – 80, GBP/INR – 100. Calculate the number of index future firm should buy to gain the above exposure. **(4)**

The following additional information is now available.

- USD/INR – 78, GBP/INR – 98 after 3 months.
 - Bid offer spread of immediate transaction – 1% for each transaction.
 - Market movement INDIA: 2% , US:2% UK:0% in 3 months.
 - Alpha generated for each market India: 4% US:0% UK:1%.
- iv)** Estimate portfolio value in 3 months if CIO goes for immediate buy and sell of equities. **(5)**
 - v)** Estimate portfolio value in 3 months where portfolio allocation is done using future instead of immediate transaction and estimate profit or loss using this methodology. **(5)**
- [20]**

- Q. 3)**
- i)** What is convexity of bond. Describe a scenario in which convexity would be a more appropriate measure of interest rate risk than modified duration. **(3)**
 - ii)** Discuss the relationship between convexity and price volatility. Explain how convexity can be used to hedge against interest rate risk. **(2)**
 - iii)** How convexity adjustment along with Modified duration is used to estimate the change in market value of bond with changes in yield. **(2)**

Two financial institutions are considering a merger to reduce the overall exposure to change in interest rate on the annual valuations. As an Investment Actuary you have been requested to provide an expert view on whether the merger would achieve the stated objective. You have been provided with the below information for your assessment.

	Institution 1		Institution 2	
	Asset	Liability	Asset	liabilities
Value	INR 100 CR	INR 100 CR	INR 100 CR	INR 100 CR
Macaulay duration	6 yr	12 yr	9 yr	3 yr
GRY	5%	8%	8%	5%
Convexity	100	200	100	50

iv) Use effective duration to discuss efficacy of merger in interest rate 2% up and 2% down scenario (round the movement to nearest Cr). (4)

v) Use both convexity and effective duration to assess efficacy of merger in interest rate 2% up and 2% down scenario (round the movement to nearest Cr) and comment which approach would be appropriate for the assessment. (4)

[15]

Q. 4)

i) How do you extend the Vasicek model to CIR model and the Hull-White model? (3)

ii) You recently attended a seminar on interest rate derivatives where the lecturer mentioned, “The Vasicek model may be appropriate for valuing interest rate swaps, while the CIR model may be better suited for valuing interest rate caps and floors. The Hull-White model may be used to model the term structure of interest rates and to value complex interest rate derivatives that depend on multiple factors.” Why do you think the lecturer made this statement? (5)

[8]

Q. 5)

i) What are synthetic CDO and how are they different from CLO. (3)

ii) Discuss relevance of attachment point in a synthetic CDO for investors. (3)

[6]

Q. 6)

You work in the risk monitoring and surveillance department of the Central Bank and have been deputed to assess a portfolio containing a number of different types of options in a scheduled commercial bank and file a report based on your findings with respect to risks that the portfolio, and thus the bank is exposed to. Upon examination, you find that the main part of the portfolio consists of vanilla currency, equity index and interest rate options. Further, the scheduled commercial bank in question tells you that the options are valued using appropriate variants of Black – Scholes option formula. The portfolio also contains underlying cash positions, plus futures and options on futures.

You also notice that a small percentage of the portfolio consists of exotic interest rate options and structured bonds, including such features as barriers and American expiries. You are also told that these products are valued using a Vasicek model.

- i) As a part of your report, describe in detail the basic market risks that would apply to this portfolio, noting especially how the risks of the options on futures and exotic options would differ from those of the vanilla options. (10)
- ii) Further, discuss the advantages and disadvantages of moving to a Hull-White model of the interest rate yield curve for this portfolio. (5)
- [15]

Q. 7) Consider the following homogeneous reference portfolio in a bank CDS:

- The number of reference entities = 100.
- CDS spread, $s = 150$ basis points.
- Recovery rate, $f = 50\%$.
- Also, assume that the defaults are all independent of each other. On a single name, the annual default probability is constant over five years and obeys the relation $s = (1 - f)PD$, where PD is the probability of default.

- i) What is the expected number of defaulting entities over the next five years? (5)
- ii) Further, which of the following tranches would be entirely wiped out (lose 100% of the principal invested) by the expected number of defaulting entities? (2)
- [7]

Q. 8) A fund manager purchases a payer swaption with option terms of 15 years and swap tenor of 5 years with a swap strike rate of 4%, a principal amount of \$10 billion priced at \$420 million. The zero – coupon bond prices for the interest rate are specified as:

Base	Zero Coupon Bond Prices
1	0.99
2	0.97
3	0.96
4	0.94
5	0.92
6	0.90
7	0.87
8	0.85
9	0.82
10	0.79
11	0.76
12	0.74
13	0.70
14	0.68
15	0.65
16	0.62
17	0.59
18	0.56
19	0.53
20	0.49

- i) Define moneyness of option. State whether the above option is in the money, at the money or out of money. (3)
- ii) Estimate the implied volatility of the swaption with confidence interval of $\pm 2\%$. Market implied volatility of 15-year swaption is between 20% - 30%. (4)
- iii) Market yield curve has changed and the new zero – coupon bond prices are as given below. Estimate the new price for this swaption. (5)

Base	Zero Coupon Bond Prices
1	1.00
2	1.00
3	0.99
4	0.98
5	0.96
6	0.94
7	0.91
8	0.88
9	0.84
10	0.81
11	0.76
12	0.72
13	0.68
14	0.64
15	0.60
16	0.55
17	0.51
18	0.46
19	0.43
20	0.38

- iv) The fund manager has priced the option after change in market curve to be \$1 billion. What are the principle causes for difference (if any) between fund manager price and your price and how can you allow for this difference in your model. (3)
- [15]

- Q. 9)** In a developing economy the Government levies Security Transaction Tax (STT) at 0.01% of contract value on futures and 0.05% of premium on options. The derivative market has been witnessing a healthy volume growth and the revenue from STT is also increasing. The Finance Minister has now proposed to increase the STT by 25% to shore up the revenue from STT. What can be the implications of this on the market? [5]
